

Estimating Phylogenetic Multilevel Models with brms

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Introduction

In the present vignette, we want to discuss how to specify phylogenetic multilevel models using **brms**. These models are relevant in evolutionary biology when data of many species are analyzed at the same time. The usual approach would be to model species as a grouping factor in a multilevel model and estimate varying intercepts (and possibly also varying slopes) over species. However, species are not independent as they come from the same phylogenetic tree and we thus have to adjust our model to incorporate this dependency. The examples discussed here are from chapter 11 of the book *Modern Phylogenetic Comparative Methods and the application in Evolutionary Biology* (de Villemereuil & Nakagawa, 2014). The necessary data can be downloaded from the corresponding website (<https://www.mpcm-evolution.com/>). Some of these models may take a few minutes to fit.

A Simple Phylogenetic Model

Assume we have measurements of a phenotype, `phen` (say the body size), and a `cofactor` variable (say the temperature of the environment). We prepare the data using the following code.

```
phylo <- ape::read.nexus("https://paul-buerkner.github.io/data/phylo.nex")
data_simple <- read.table(
  "https://paul-buerkner.github.io/data/data_simple.txt",
  header = TRUE
)
head(data_simple)
```

```
      phen  cofactor phylo
1 107.06595 10.309588 sp_1
2  79.61086  9.690507 sp_2
3 116.38186 15.007825 sp_3
4 143.28705 19.087673 sp_4
```

```
5 139.60993 15.658404 sp_5
6 68.50657 6.005236 sp_6
```

The `phylo` object contains information on the relationship between species. Using this information, we can construct a covariance matrix of species (Hadfield & Nakagawa, 2010).

```
A <- ape::vcv.phylo(phylo)
```

Now we are ready to fit our first phylogenetic multilevel model:

```
model_simple <- brm(
  phen ~ cofactor + (1|gr(phylo, cov = A)),
  data = data_simple,
  family = gaussian(),
  data2 = list(A = A),
  prior = c(
    prior(normal(0, 10), "b"),
    prior(normal(0, 50), "Intercept"),
    prior(student_t(3, 0, 20), "sd"),
    prior(student_t(3, 0, 20), "sigma")
  )
)
```

With the exception of `(1|gr(phylo, cov = A))` instead of `(1|phylo)` this is a basic multilevel model with a varying intercept over species (`phylo` is an indicator of species in this data set). However, by using `cov = A` in the `gr` function, we make sure that species are correlated as specified by the covariance matrix `A`. We pass `A` itself via the `data2` argument which can be used for any kinds of data that does not fit into the regular structure of the `data` argument. Setting priors is not required for achieving good convergence for this model, but it improves sampling speed a bit. After fitting, the results can be investigated in detail.

```
summary(model_simple)
```

```
Family: gaussian
Links: mu = identity
Formula: phen ~ cofactor + (1 | gr(phylo, cov = A))
Data: data_simple (Number of observations: 200)
Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
       total post-warmup draws = 4000
```

Multilevel Hyperparameters:

```
~phylo (Number of levels: 200)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)   14.38     2.10   10.43   18.62 1.00     987    1754
```

Regression Coefficients:

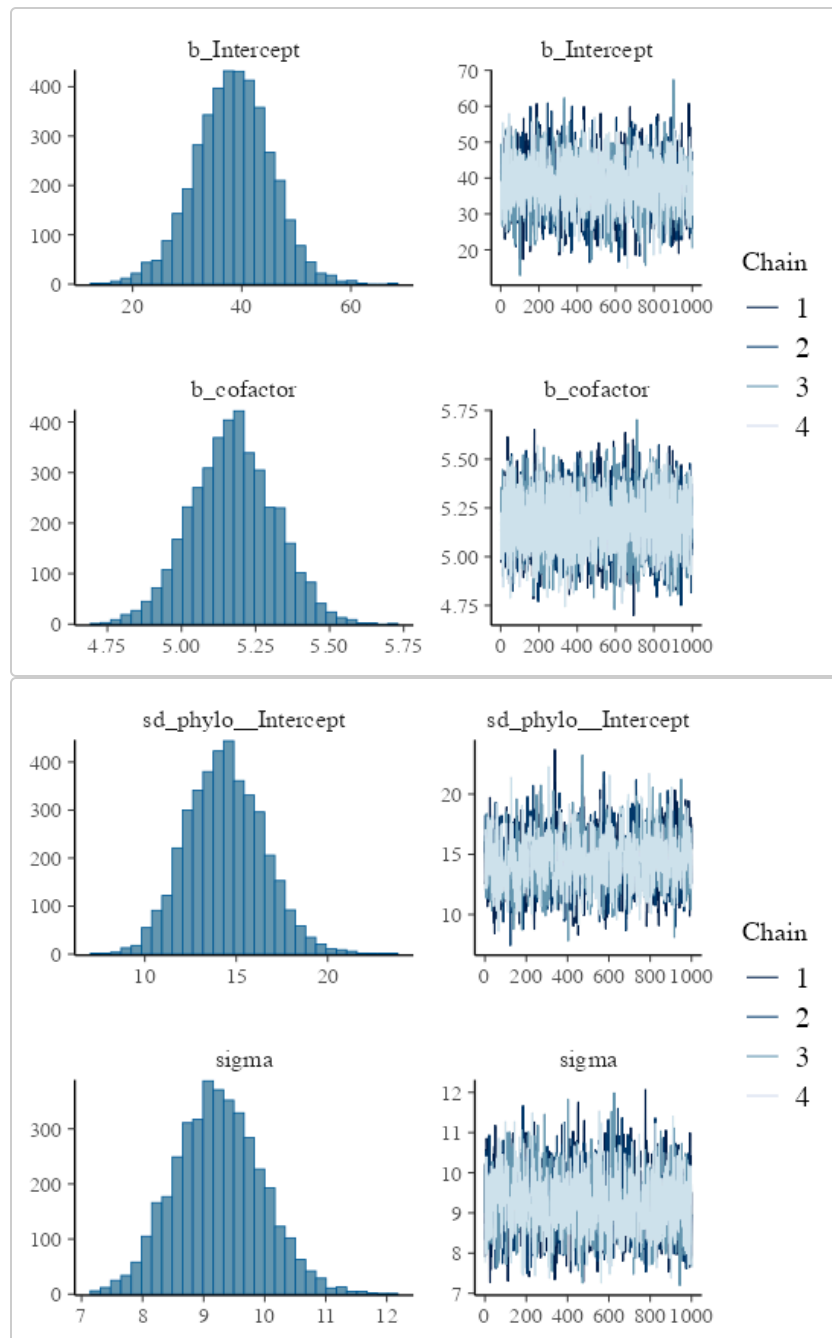
```
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept   38.38     6.92   24.09   51.70 1.00     2085    2633
cofactor     5.17     0.14    4.90    5.45 1.00     6236    3311
```

Further Distributional Parameters:

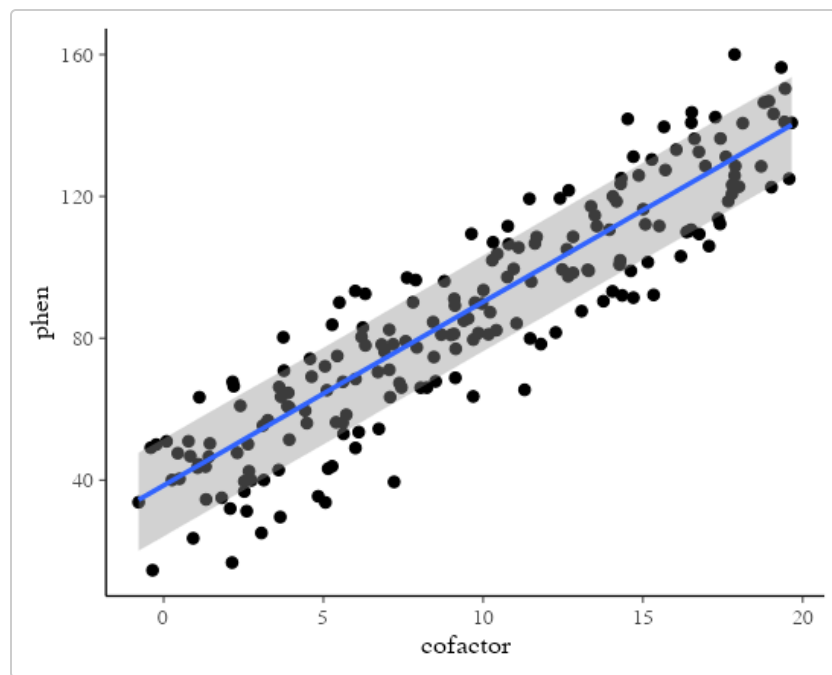
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	9.24	0.72	7.91	10.71	1.00	1364	2240

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(model_simple, N = 2, ask = FALSE)
```



```
plot(conditional_effects(model_simple), points = TRUE)
```



The so called phylogenetic signal (often symbolize by λ) can be computed with the `hypothesis` method and is roughly $\lambda = 0.7$ for this example.

```
hyp <- "sd_phylo__Intercept^2 / (sd_phylo__Intercept^2 + sigma^2) = 0"
(hyp <- hypothesis(model_simple, hyp, class = NULL))
```

Hypothesis Tests for class :

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1	(sd_phylo__Interc... = 0	0.7	0.08	0.51	0.83	NA	NA	*

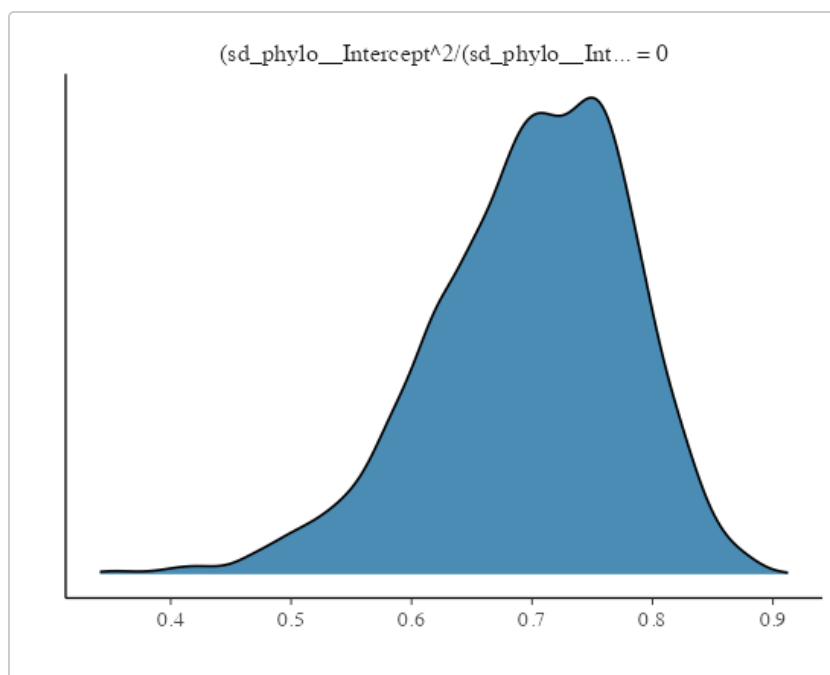
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.

'*': For one-sided hypotheses, the posterior probability exceeds 95%;

for two-sided hypotheses, the value tested against lies outside the 95%-CI.

Posterior probabilities of point hypotheses assume equal prior probabilities.

```
plot(hyp)
```



Note that the phylogenetic signal is just a synonym of the intra-class correlation (ICC) used in the context phylogenetic analysis.

A Phylogenetic Model with Repeated Measurements

Often, we have multiple observations per species and this allows to fit more complicated phylogenetic models.

```
data_repeat <- read.table(
  "https://paul-buerkner.github.io/data/data_repeat.txt",
  header = TRUE
)
data_repeat$spec_mean_cf <-
  with(data_repeat, sapply(split(cofactor, phylo), mean)[phylo])
head(data_repeat)
```

	phen	cofactor	species	phylo	spec_mean_cf
1	107.41919	11.223724	sp_1	sp_1	10.309588
2	109.16403	9.805934	sp_1	sp_1	10.309588
3	91.88672	10.308423	sp_1	sp_1	10.309588
4	121.54341	8.355349	sp_1	sp_1	10.309588
5	105.31638	11.854510	sp_1	sp_1	10.309588
6	64.99859	4.314015	sp_2	sp_2	3.673914

The variable `spec_mean_cf` just contains the mean of the cofactor for each species. The code for the repeated measurement phylogenetic model looks as follows:

```
model_repeat1 <- brm(
  phen ~ spec_mean_cf + (1|gr(phylo, cov = A)) + (1|species),
  data = data_repeat,
  family = gaussian(),
  data2 = list(A = A),
```

```
prior = c(
  prior(normal(0,10), "b"),
  prior(normal(0,50), "Intercept"),
  prior(student_t(3,0,20), "sd"),
  prior(student_t(3,0,20), "sigma")
),
sample_prior = TRUE, chains = 2, cores = 2,
iter = 4000, warmup = 1000
)
```

The variables `phylo` and `species` are identical as they are both identifiers of the species. However, we model the phylogenetic covariance only for `phylo` and thus the `species` variable accounts for any specific effect that would be independent of the phylogenetic relationship between species (e.g., environmental or niche effects). Again we can obtain model summaries as well as estimates of the phylogenetic signal.

```
summary(model_repeat1)
```

```
Family: gaussian
Links: mu = identity
Formula: phen ~ spec_mean_cf + (1 | gr(phylo, cov = A)) + (1 | species)
Data: data_repeat (Number of observations: 1000)
Draws: 2 chains, each with iter = 4000; warmup = 1000; thin = 1;
       total post-warmup draws = 6000
```

Multilevel Hyperparameters:

```
~phylo (Number of levels: 200)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)   16.48     1.93   12.89   20.54 1.00    1209    1927
```

```
~species (Number of levels: 200)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    4.96     0.86    3.18    6.56 1.00     993    1293
```

Regression Coefficients:

```
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    35.94     7.92   20.59   51.41 1.00    3262    3487
spec_mean_cf   5.10     0.11    4.89    5.31 1.00    6767    4285
```

Further Distributional Parameters:

```
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     8.11     0.20    7.72    8.52 1.00    5406    3733
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
hyp <- paste(
  "sd_phylo__Intercept^2 /",
  "(sd_phylo__Intercept^2 + sd_species__Intercept^2 + sigma^2) = 0"
```

```
)
(hyp <- hypothesis(model_repeat1, hyp, class = NULL))
```

Hypothesis Tests for class :

	Hypothesis	Estimate	Est.Error	CI.Lower	CI.Upper	Evid.Ratio	Post.Prob	Star
1	(sd_phylo__Intercept... = 0	0.74	0.06	0.62	0.84	0	0	*

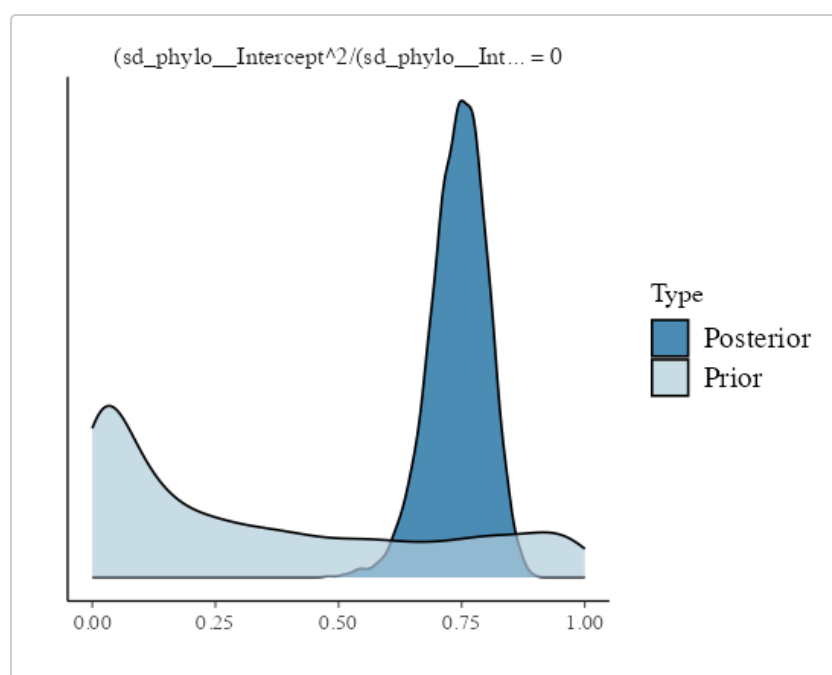
'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.

'*': For one-sided hypotheses, the posterior probability exceeds 95%;

for two-sided hypotheses, the value tested against lies outside the 95%-CI.

Posterior probabilities of point hypotheses assume equal prior probabilities.

```
plot(hyp)
```



So far, we have completely ignored the variability of the cofactor within species. To incorporate this into the model, we define

```
data_repeat$within_spec_cf <- data_repeat$cofactor - data_repeat$spec_mean_cf
```

and then fit it again using `within_spec_cf` as an additional predictor.

```
model_repeat2 <- update(
  model_repeat1, formula = ~ . + within_spec_cf,
  newdata = data_repeat, chains = 2, cores = 2,
  iter = 4000, warmup = 1000
)
```

The results are almost unchanged, with apparently no relationship between the phenotype and the within species variance of cofactor.

```
summary(model_repeat2)
```

```
Family: gaussian
Links: mu = identity
Formula: phen ~ spec_mean_cf + (1 | gr(phylo, cov = A)) + (1 | species) + within_spec_cf
Data: data_repeat (Number of observations: 1000)
Draws: 2 chains, each with iter = 4000; warmup = 1000; thin = 1;
       total post-warmup draws = 6000
```

Multilevel Hyperparameters:

```
~phylo (Number of levels: 200)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)   16.38     1.86   12.99   20.21 1.00    1114    2039
```

```
~species (Number of levels: 200)
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    4.98     0.83    3.36    6.67 1.00     967    1608
```

Regression Coefficients:

```
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept      36.13     7.89   20.64   51.62 1.00    2595    3088
spec_mean_cf    5.10     0.10    4.89    5.31 1.00    5425    4374
within_spec_cf  -0.06     0.19   -0.43    0.30 1.00   11877    4202
```

Further Distributional Parameters:

```
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma      8.11     0.20    7.73    8.53 1.00    6005    4372
```

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

Also, the phylogenetic signal remains more or less the same.

```
hyp <- paste(
  "sd_phylo__Intercept^2 /",
  "(sd_phylo__Intercept^2 + sd_species__Intercept^2 + sigma^2) = 0"
)
(hyp <- hypothesis(model_repeat2, hyp, class = NULL))
```

Hypothesis Tests for class :

```
      Hypothesis Estimate Est.Error CI.Lower CI.Upper Evid.Ratio Post.Prob Star
1 (sd_phylo__Interc... = 0      0.74      0.06    0.62    0.83          0          0    *
---
```

'CI': 90%-CI for one-sided and 95%-CI for two-sided hypotheses.

'*': For one-sided hypotheses, the posterior probability exceeds 95%;

for two-sided hypotheses, the value tested against lies outside the 95%-CI.

Posterior probabilities of point hypotheses assume equal prior probabilities.

A Phylogenetic Meta-Analysis

Let's say we have Fisher's z-transformed correlation coefficients Zr per species along with corresponding sample sizes (e.g., correlations between male coloration and reproductive success):

```
data_fisher <- read.table(
  "https://paul-buerkner.github.io/data/data_effect.txt",
  header = TRUE
)
data_fisher$obs <- 1:nrow(data_fisher)
head(data_fisher)
```

```
      Zr   N phylo obs
1 0.28917549 13  sp_1  1
2 0.02415579 40  sp_2  2
3 0.19513651 39  sp_3  3
4 0.09831239 40  sp_4  4
5 0.13780152 66  sp_5  5
6 0.13710587 41  sp_6  6
```

We assume the sampling variance to be known and as $V(Zr) = \frac{1}{N-3}$ for Fisher's values, where N is the sample size per species. Incorporating the known sampling variance into the model is straight forward. One has to keep in mind though, that **brms** requires the sampling standard deviation (square root of the variance) as input instead of the variance itself. The group-level effect of `obs` represents the residual variance, which we have to model explicitly in a meta-analytic model.

```
model_fisher <- brm(
  Zr | se(sqrt(1 / (N - 3))) ~ 1 + (1|gr(phylo, cov = A)) + (1|obs),
  data = data_fisher, family = gaussian(),
  data2 = list(A = A),
  prior = c(
    prior(normal(0, 10), "Intercept"),
    prior(student_t(3, 0, 10), "sd")
  ),
  control = list(adapt_delta = 0.95),
  chains = 2, cores = 2, iter = 4000, warmup = 1000
)
```

A summary of the fitted model is obtained via

```
summary(model_fisher)
```

```
Family: gaussian
Links: mu = identity
Formula: Zr | se(sqrt(1/(N - 3))) ~ 1 + (1 | gr(phylo, cov = A)) + (1 | obs)
Data: data_fisher (Number of observations: 200)
Draws: 2 chains, each with iter = 4000; warmup = 1000; thin = 1;
```

total post-warmup draws = 6000

Multilevel Hyperparameters:

~obs (Number of levels: 200)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.05	0.03	0.00	0.10	1.00	751	1293

~phylo (Number of levels: 200)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.07	0.04	0.00	0.15	1.00	751	1498

Regression Coefficients:

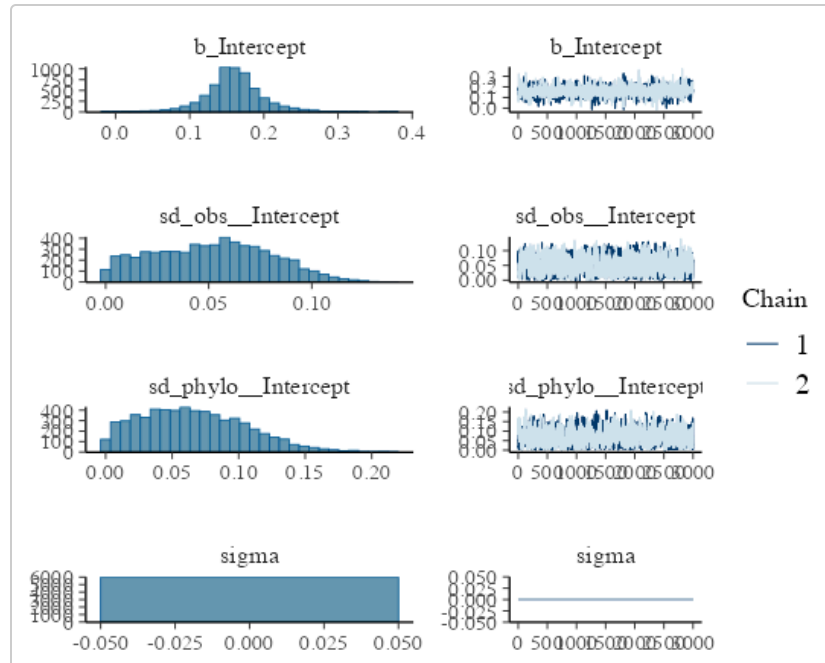
	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	0.16	0.04	0.08	0.25	1.00	2692	2344

Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	0.00	0.00	0.00	0.00	NA	NA	NA

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

`plot(model_fisher)`



The meta-analytic mean (i.e., the model intercept) is 0.16 with a credible interval of [0.08, 0.25]. Thus the mean correlation across species is positive according to the model.

A phylogenetic count-data model

Suppose that we analyze a phenotype that consists of counts instead of being a continuous variable. In such a case, the normality assumption will likely not be justified and it is recommended to use a distribution explicitly suited for count data, for instance the Poisson distribution. The following data set (again retrieved from mpcm-evolution.org) provides an example.

```
data_pois <- read.table(
  "https://paul-buerkner.github.io/data/data_pois.txt",
  header = TRUE
)
data_pois$obs <- 1:nrow(data_pois)
head(data_pois)
```

	phen_pois	cofactor	phylo	obs
1	1	7.8702830	sp_1	1
2	0	3.4690529	sp_2	2
3	1	2.5478774	sp_3	3
4	14	18.2286628	sp_4	4
5	1	2.5302806	sp_5	5
6	1	0.5145559	sp_6	6

As the Poisson distribution does not have a natural overdispersion parameter, we model the residual variance via the group-level effects of `obs` (e.g., see Lawless, 1987).

```
model_pois <- brm(
  phen_pois ~ cofactor + (1|gr(phylo, cov = A)) + (1|obs),
  data = data_pois, family = poisson("log"),
  data2 = list(A = A),
  chains = 2, cores = 2, iter = 4000,
  control = list(adapt_delta = 0.95)
)
```

Again, we obtain a summary of the fitted model via

```
summary(model_pois)
```

```
Family: poisson
Links: mu = log
Formula: phen_pois ~ cofactor + (1 | gr(phylo, cov = A)) + (1 | obs)
Data: data_pois (Number of observations: 200)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
       total post-warmup draws = 4000

Multilevel Hyperparameters:
~obs (Number of levels: 200)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.19    0.08    0.02    0.34 1.00    513    509

~phylo (Number of levels: 200)
      Estimate Est.Error 1-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
```

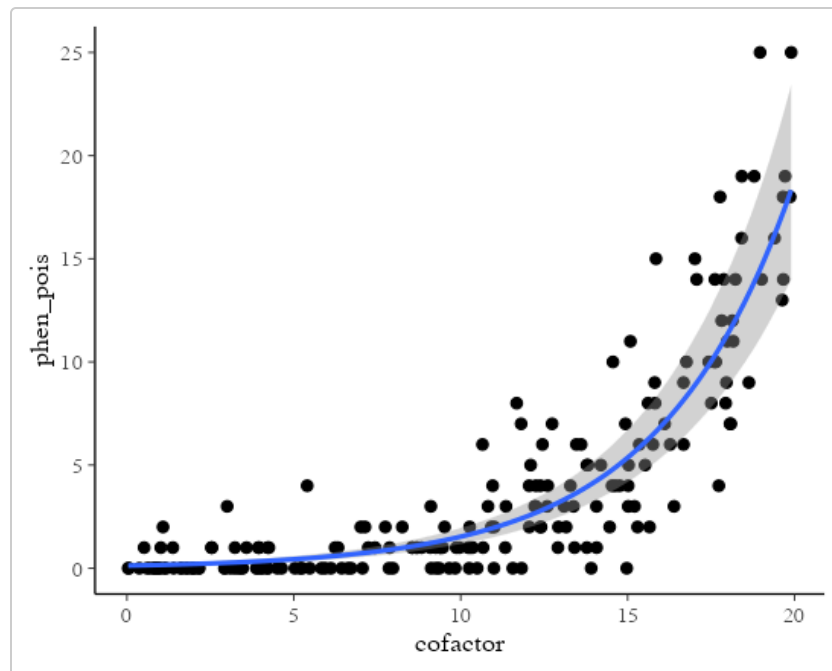
```
sd(Intercept)    0.18    0.10    0.02    0.42 1.01    584    1188
```

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-2.09	0.20	-2.49	-1.70	1.00	2157	2357
cofactor	0.25	0.01	0.23	0.27	1.00	3466	3038

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

```
plot(conditional_effects(model_pois), points = TRUE)
```



Now, assume we ignore the fact that the phenotype is count data and fit a linear normal model instead.

```
model_normal <- brm(
  phen_pois ~ cofactor + (1|gr(phylo, cov = A)),
  data = data_pois, family = gaussian(),
  data2 = list(A = A),
  chains = 2, cores = 2, iter = 4000,
  control = list(adapt_delta = 0.95)
)
```

```
summary(model_normal)
```

```
Family: gaussian
Links: mu = identity
Formula: phen_pois ~ cofactor + (1 | gr(phylo, cov = A))
Data: data_pois (Number of observations: 200)
Draws: 2 chains, each with iter = 4000; warmup = 2000; thin = 1;
```

total post-warmup draws = 4000

Multilevel Hyperparameters:

~phylo (Number of levels: 200)

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sd(Intercept)	0.67	0.48	0.03	1.78	1.00	1264	2013

Regression Coefficients:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
Intercept	-3.05	0.61	-4.31	-1.88	1.00	4559	2783
cofactor	0.68	0.04	0.60	0.76	1.00	7806	2528

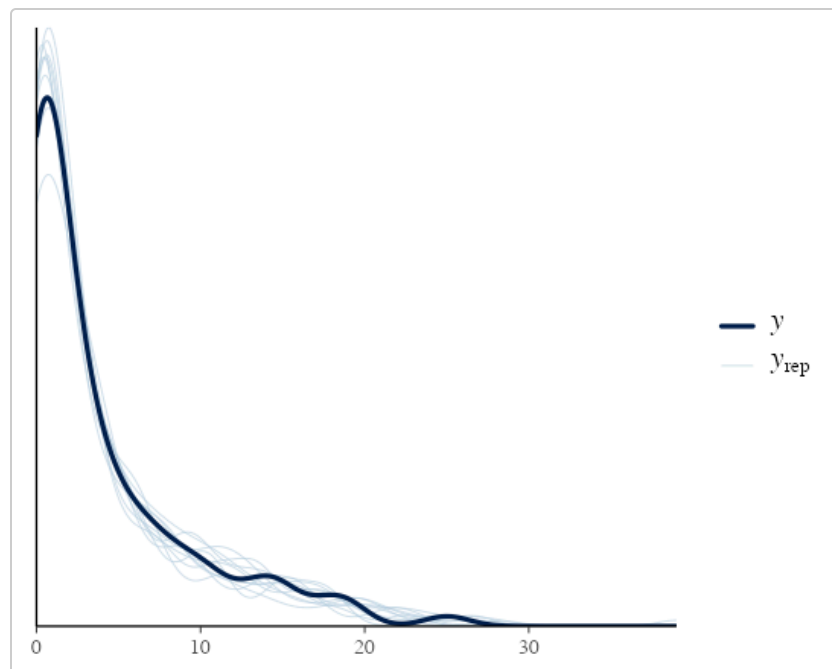
Further Distributional Parameters:

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
sigma	3.44	0.18	3.11	3.81	1.00	5621	2727

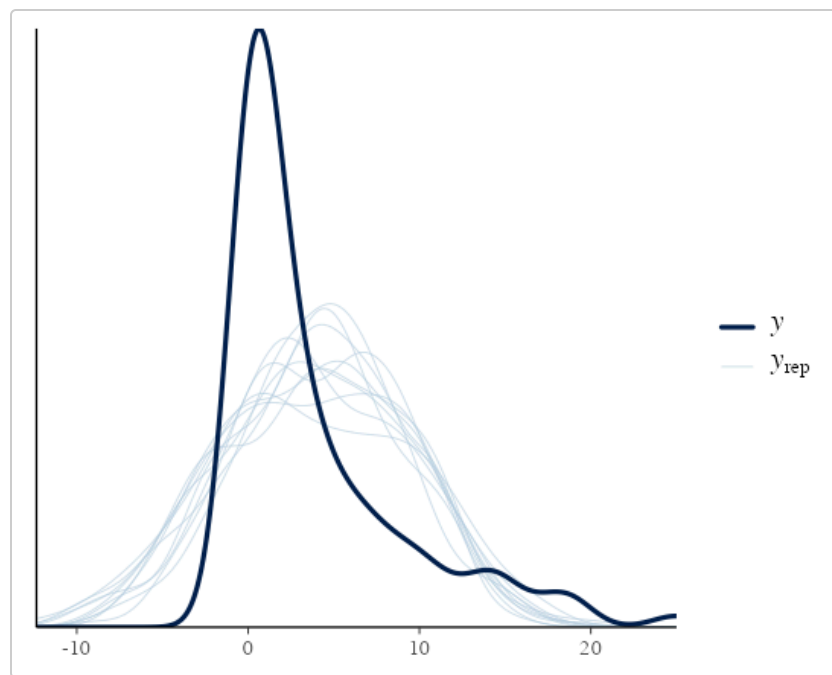
Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

We see that `cofactor` has a positive relationship with the phenotype in both models. One should keep in mind, though, that the estimates of the Poisson model are on the log-scale, as we applied the canonical log-link function in this example. Therefore, estimates are not comparable to a linear normal model even if applied to the same data. What we can compare, however, is the model fit, for instance graphically via posterior predictive checks.

```
pp_check(model_pois)
```



```
pp_check(model_normal)
```



Apparently, the distribution of the phenotype predicted by the Poisson model resembles the original distribution of the phenotype pretty closely, while the normal models fails to do so. We can also apply leave-one-out cross-validation for direct numerical comparison of model fit.

```
loo(model_pois, model_normal)
```

Output of model 'model_pois':

Computed from 4000 by 200 log-likelihood matrix.

	Estimate	SE
elpd_loo	-348.1	17.0
p_loo	30.7	3.5
looic	696.2	33.9

MCSE of elpd_loo is NA.

MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.4]).

Pareto k diagnostic values:

		Count	Pct.	Min.	ESS
(-Inf, 0.7]	(good)	195	97.5%	253	
(0.7, 1]	(bad)	5	2.5%	<NA>	
(1, Inf)	(very bad)	0	0.0%	<NA>	

See help('pareto-k-diagnostic') for details.

Output of model 'model_normal':

Computed from 4000 by 200 log-likelihood matrix.

	Estimate	SE
elpd_loo	-535.9	15.9
p_loo	9.8	2.2

```
looic      1071.9 31.7
```

```
-----
```

```
MCSE of elpd_loo is 0.1.
```

```
MCSE and ESS estimates assume MCMC draws (r_eff in [0.6, 1.8]).
```

```
All Pareto k estimates are good (k < 0.7).
```

```
See help('pareto-k-diagnostic') for details.
```

```
Model comparisons:
```

```
      elpd_diff se_diff
model_pois      0.0      0.0
model_normal -187.8     18.0
```

Since smaller values of loo indicate better fit, it is again evident that the Poisson model fits the data better than the normal model. Of course, the Poisson model is not the only reasonable option here. For instance, you could use a negative binomial model (via family `negative_binomial`), which already contains an overdispersion parameter so that modeling a varying intercept of `obs` becomes obsolete.

Phylogenetic models with multiple group-level effects

In the above examples, we have only used a single group-level effect (i.e., a varying intercept) for the phylogenetic grouping factors. In **brms**, it is also possible to estimate multiple group-level effects (e.g., a varying intercept and a varying slope) for these grouping factors. However, it requires repeatedly computing Kronecker products of covariance matrices while fitting the model. This will be very slow especially when the grouping factors have many levels and matrices are thus large.

References

- de Villemeruil P. & Nakagawa, S. (2014) General quantitative genetic methods for comparative biology. In: *Modern phylogenetic comparative methods and their application in evolutionary biology: concepts and practice* (ed. Garamszegi L.) Springer, New York. pp. 287-303.
- Hadfield, J. D. & Nakagawa, S. (2010) General quantitative genetic methods for comparative biology: phylogenies, taxonomies, and multi-trait models for continuous and categorical characters. *Journal of Evolutionary Biology*. 23. 494-508.
- Lawless, J. F. (1987). Negative binomial and mixed Poisson regression. *Canadian Journal of Statistics*, 15(3), 209-225.